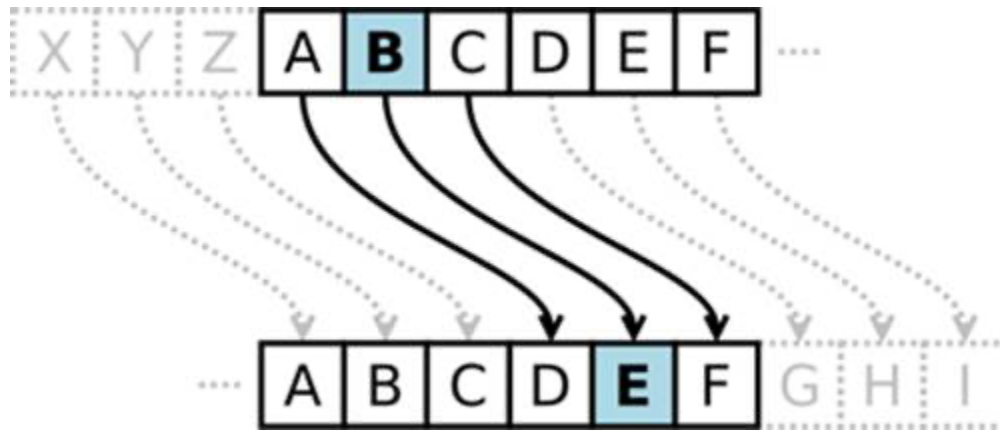


APS106 Design Problem

Week 7: Cryptography



Problem Background

You are a student living in residence and you have the best gaming computer. As a result, many of your friends keep asking to use your computer when you are not around, which you don't mind. However, you have a lot of sensitive information that are for your eyes only. You want to be a good friend, but you also want to keep your information safe. You decide that the option that's best for everyone is to encrypt all your sensitive notes, that way only you will be able to read them. After doing some reading you realize that there is a field called *cryptography* that focuses on the science of writing secret codes.

For thousands of years, cryptography has studied the creation of secret messages that only the sender and recipient could read. The secret code system (or algorithm) used for encrypting and decrypting messages is called a *cipher*. Even if the messenger were to be captured, it would not help in decoding the message, since only sender and recipient would know the cipher. More precisely everyone might know the algorithm but the key to the cipher is only known to the sender and receiver.

One early example of cryptography is the Caesar cipher that encrypts a message by taking each letter in the message and replacing it with a "shifted" letter. If you shift the letter A by one space, you get the letter B. If you shift the letter A by two spaces, you get the letter C (see the figure above). The Caesar cipher is overly simple and it wouldn't take much to break the encryption. Even though the Caesar cipher is no longer used, it still makes for a great learning example for cryptography.

1. Define the Problem

Given a sensitive message that should not be read by others, we would like to devise an algorithm to encrypt the message so that only someone with the secret key can access it.

There are many options available for encryption ranging from classic methods that are not secure, to more modern ones that are highly mathematical. Since the message being encrypted is not of great importance, and there is little motivation for your friends to try and decrypt it, it makes sense to use an easier implementation, such as the Caesar cipher.

More precisely, you should develop code to do two things using the Caesar cipher:

1. Prompt the user for a message, a secret key, and whether you want the message encoded or decoded.
2. Encode or decode the message using a Caesar cipher and it print out.

2. Define Test Cases

Create a few secret messages and their corresponding encrypted text.

3. Generate Many Creative Solutions

How are you going to break down the steps in this algorithm? Create an Algorithm Plan.

Most of the steps should be easy but the actual encryption/decryption is (a bit) harder. What do you need to do to implement the Caesar cipher? How can you shift letters? Write some code with one of your test cases to figure out if you can print out the encrypted text.

4. Select a Solution

Does the best solution you figured out in Step 3 make sense?

5. Implement the Solution

Program it step-by-step. What functionality should go into functions?

Test each step as you go.

6. Perform Final Testing

Run all of our tests to make sure it is working. Run some more. Try to come up with strange inputs that will break your code and then fix it.